

# A tour of the model zoo

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## 1 Overview

`EcologicalBayesianNetworks.jl` is a curated zoo of publicly downloadable ecological Bayesian belief networks and influence diagrams. Every model has a manifest (`models/<name>/metadata.toml`) recording where it came from, how to cite it, its licence and a checksum. A permissive licence (CC BY, CC BY-SA, CC0, MIT) is what *permits* shipping the original file with the package, but it does not by itself decide: each manifest's `redistribution` field does, and several permissively licensed records are still `fetch-only` or `reconstruction` because of how the source publishes them. Records that are not shipped verbatim are fetched into a local cache on demand. This vignette walks through the catalogue, loads a benchmark network, solves a decision model, and computes a marginal on a small chance-only network.

The models are real ones: belief networks published for habitat, conservation and natural-resource problems (McCann et al. 2006; Newton et al. 2007), environmental risk assessments (Moe et al. 2021; Kaikkonen et al. 2021) and ecosystem-service models (Landuyt et al. 2013). They come from four kinds of source – the Bayesian Network Model Archive (BNMA 2026), the bnlearn network repository (Scutari 2010), the bnRep collection (Leonelli 2025), and the supplementary material of individual papers – and each brings its own licence, which is why the zoo enforces a redistribution policy rather than mirroring everything.

## 2 Setup

```
using EcologicalBayesianNetworks
n_models()
```

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## 3 The catalogue

`model_catalog()` returns a filtered view of the registry that prints as a table. The `kind` column says whether a model is a plain Bayesian network (BN), has decision or utility nodes (ID, an influence diagram), or is a package: a catalogue record for a Dryad, Zenodo, figshare or source-repository archive from which a network can be rebuilt but which is not itself a network file.

```
model_catalog()
```

ModelCatalog with 41 models

| name                                    | category  | format     | licence                       |
|-----------------------------------------|-----------|------------|-------------------------------|
| redistribution nodes arcs kind          |           |            |                               |
| reference_grazing_id                    | reference | julia      | MIT                           |
| builtin 12 13 ID                        |           |            |                               |
| reference_habitat_bn                    | reference | julia      | MIT                           |
| builtin 7 6 BN                          |           |            |                               |
| beach_mice_bn                           | habitat   | neta       | CC0 1.0 (ScienceBase record); |
| US Government work in the public domain |           | fetch-only | 0 0 BN                        |
| brown_bear_recreation                   | habitat   | dne        | CC-BY-NC-ND (version          |
| unstated)                               |           |            | fetch-only 37 45              |
| BN                                      |           |            |                               |
| gulf_sturgeon_bn                        | habitat   | neta       | CC0 1.0 (ScienceBase record); |
| US Government work in the public domain |           | fetch-only | 0 0 BN                        |
| habitat_suitability_tiger               | habitat   | xdsl       | CC-BY (version unstated)      |
| verbatim 11 12 BN                       |           |            |                               |

|                                 |    |    |             |         |                                                                              |                |       |
|---------------------------------|----|----|-------------|---------|------------------------------------------------------------------------------|----------------|-------|
| plexus_teb_bat_site             |    |    | habitat     | dne     | unstated                                                                     |                |       |
| fetch-only                      | 13 | 15 | BN          |         |                                                                              |                |       |
| plexus_teb_bat_subwatershed     |    |    | habitat     | dne     | unstated                                                                     |                |       |
| fetch-only                      | 7  | 6  | BN          |         |                                                                              |                |       |
| seagrass_dbn_package            |    |    | habitat     | package | Dryad record terms (consult the record; Dryad lists its content as reusable) | reconstruction | 0 0   |
| package                         |    |    |             |         |                                                                              |                |       |
| thailand_vertebrate_bbn_package |    |    | habitat     | package | figshare collection terms                                                    | reconstruction | 0 0   |
| (consult the individual items)  |    |    |             |         |                                                                              |                |       |
| package                         |    |    |             |         |                                                                              |                |       |
| bull_trout_food_web             |    |    | population  | dne     | CC-BY-NC-ND (version unstated)                                               | fetch-only     | 10 15 |
| BN                              |    |    |             |         |                                                                              |                |       |
| marten_age                      |    |    | population  | dne     | CC-BY (version unstated)                                                     |                |       |
| fetch-only                      | 7  | 6  | BN          |         |                                                                              |                |       |
| marten_telomere                 |    |    | population  | dne     | CC-BY-NC-ND (version unstated)                                               | fetch-only     | 0 0   |
| BN                              |    |    |             |         |                                                                              |                |       |
| native_fish_v1                  |    |    | population  | dne     | CC-BY (version unstated)                                                     |                |       |
| verbatim                        | 7  | 8  | BN          |         |                                                                              |                |       |
| pacific_walrus                  |    |    | population  | dne     | CC-BY-NC-ND (version unstated)                                               | fetch-only     | 77 96 |
| BN                              |    |    |             |         |                                                                              |                |       |
| plexus_population_viability     |    |    | population  | dne     | unstated                                                                     |                |       |
| fetch-only                      | 11 | 13 | ID          |         |                                                                              |                |       |
| plexus_teb_bat_basin            |    |    | population  | dne     | unstated                                                                     |                |       |
| fetch-only                      | 8  | 7  | BN          |         |                                                                              |                |       |
| polar_bear_stressor_i           |    |    | population  | dne     | CC-BY-NC-ND (version unstated)                                               | fetch-only     | 38 44 |
| BN                              |    |    |             |         |                                                                              |                |       |
| polar_bear_stressor_ii          |    |    | population  | dne     | CC-BY-NC-ND (version unstated)                                               | fetch-only     | 49 53 |
| BN                              |    |    |             |         |                                                                              |                |       |
| ballycanew_diffuse_p            |    |    | water       | xdsl    | MIT                                                                          |                |       |
| verbatim                        | 4  | 2  | BN          |         |                                                                              |                |       |
| ballycanew_point_diffuse_p      |    |    | water       | xdsl    | MIT                                                                          |                |       |
| verbatim                        | 12 | 8  | BN          |         |                                                                              |                |       |
| bnma_water                      |    |    | water       | dne     | CC-BY-SA (version unstated)                                                  |                |       |
| verbatim                        | 32 | 66 | BN          |         |                                                                              |                |       |
| vansjo_seasonal_bn_package      |    |    | water       | package | unstated (repository states no licence; check the repository terms)          | reconstruction | 0 0   |
| package                         |    |    |             |         |                                                                              |                |       |
| shark_management_survey         |    |    | risk        | dne     | CC-BY-NC (version unstated)                                                  |                |       |
| fetch-only                      | 26 | 54 | BN          |         |                                                                              |                |       |
| tidal_saline_wetlands           |    |    | risk        | dne     | CC-BY-ND (version unstated)                                                  |                |       |
| fetch-only                      | 18 | 16 | BN          |         |                                                                              |                |       |
| arhopalus_flight_activity       |    |    | biosecurity | dne     | CC-BY-NC-ND (version unstated)                                               | fetch-only     | 7 11  |
| BN                              |    |    |             |         |                                                                              |                |       |
| fisram_freshwater_species       |    |    | biosecurity | dne     | CC-BY-ND (version unstated)                                                  |                |       |
| fetch-only                      | 20 | 23 | BN          |         |                                                                              |                |       |

|                                        |    |    |             |           |                                    |                                    |    |
|----------------------------------------|----|----|-------------|-----------|------------------------------------|------------------------------------|----|
| hylastes_flight_activity<br>unstated)  |    |    | biosecurity | dne       | CC-BY-NC-ND (version<br>fetch-only | 7                                  | 11 |
| BN                                     |    |    |             |           |                                    |                                    |    |
| hylurgus_flight_activity<br>unstated)  |    |    | biosecurity | dne       | CC-BY-NC-ND (version<br>fetch-only | 10                                 | 17 |
| BN                                     |    |    |             |           |                                    |                                    |    |
| microctonus_ng_risk<br>unstated)       |    |    | biosecurity | xdsl      | CC-BY-NC-ND (version<br>fetch-only | 45                                 | 51 |
| BN                                     |    |    |             |           |                                    |                                    |    |
| wild_pig_hsi_package<br>reconstruction | 0  | 0  | package     | package   | CC0 1.0                            |                                    |    |
| brown_trout_bdn<br>verbatim            | 9  | 13 | ID          | decision  | dne                                | CC-BY (version unstated)           |    |
| koalas<br>verbatim                     | 10 | 11 | ID          | decision  | dne                                | CC-BY (version unstated)           |    |
| song_sparrow_bdn<br>verbatim           | 12 | 15 | ID          | decision  | dne                                | CC-BY (version unstated)           |    |
| waterhole_fence<br>verbatim            | 10 | 12 | ID          | decision  | dne                                | CC-BY (version unstated)           |    |
| animals<br>unstated)                   |    |    |             | benchmark | dne                                | CC-BY-NC-ND (version<br>fetch-only | 7  |
| BN                                     |    |    |             |           |                                    |                                    | 6  |
| barley<br>verbatim                     | 48 | 84 | BN          | benchmark | net                                | CC BY-SA 3.0                       |    |
| bnrep_package<br>reconstruction        | 0  | 0  | package     | benchmark | package                            | MIT                                |    |
| hailfinder<br>verbatim                 | 56 | 66 | BN          | benchmark | net                                | CC BY-SA 3.0                       |    |
| mildew<br>verbatim                     | 35 | 46 | BN          | benchmark | net                                | CC BY-SA 3.0                       |    |
| water<br>verbatim                      | 32 | 66 | BN          | benchmark | net                                | CC BY-SA 3.0                       |    |

Categories, in display order:

```
model_categories()
```

```
8-element Vector{String}:
```

```
"reference"
"habitat"
"population"
"water"
"risk"
"biosecurity"
"decision"
"benchmark"
```

Filters combine: the benchmark networks, or only the influence diagrams whose files ship with the package.

```
available_models(category = "benchmark")
```

```
6-element Vector{String}:  
"animals"  
"barley"  
"bnrep_package"  
"hailfinder"  
"mildew"  
"water"
```

```
available_models(kind = :id, redistributable = true)
```

```
5-element Vector{String}:  
"brown_trout_bdn"  
"koalas"  
"reference_grazing_id"  
"song_sparrow_bdn"  
"waterhole_fence"
```

`model_info` looks a model up by name, title or file name, ignoring case and punctuation, and returns its `ModelSpec`.

```
model_info("Song Sparrow BDN")
```

```
ModelSpec song_sparrow_bdn  
  title          Song Sparrow riparian restoration decision network  
  (SongSparrowBDN_v250804)  
  category       decision  
  tags           bird, riparian, restoration, mercury, decision network, BNMA  
  format         dne  
  file           SongSparrowBDN_v250804.dne  
  licence        CC-BY (version unstated)  
  redistribution  verbatim  
  nodes / arcs   12 / 15  
  kind           influence diagram  
  source         https://bnma.co/bn/2893  
  download       https://bnma.co/bnrepo/?slice=download&format=dne&bnId=2893  
  doi            10.59381/ofwegrfaqyo  
  citation       Kotalik, C. J., Rowland, F. E., Marcot, B. G., Skrabis, K.,  
  Walters, D. M., Hinck, J. E., Clements, W., Richer, E. and Isanhart, J. (2026).  
  Causal networks to inform decisions for ecological restoration. Environmental  
  Management 76(7).  
  retrieved      2026-09-06  
  sha256         9be7fb3ba119d0f5c8a00509fe87d19827043b6ac26d7dce5b57c679c7181666  
  parser options allow_missing_tables = true, strict = true
```

notes                      Original .dne as served by the BNMA converter (BN DOI 10.59381/ofwegrfqyo); filename unchanged. Decision network with planting and soil-treatment decisions and utility nodes (8 chance, 2 decision, 2 utility nodes; 7 CONSTANT nodes excluded); load\_model returns an InfluenceDiagramModel whose two table-less mechanisms stay unbound, so it cannot be optimised. Two chance nodes (RiparianPlantCost\_infl\_adj, Time) are Netica finding/equation nodes with evidence but no CPT, so the manifest sets allow\_missing\_tables = true and model\_ir validates with BayesianNetworkFormats.validate(ir; allow\_missing\_tables = true).

## 4 Licences and redistribution

The manifest redistribution field is the licence policy in one word:

- verbatim: the original file is committed next to a LICENSE.txt with the attribution (the CC BY-SA 3.0 benchmarks of bnlearn ([Scutari 2010](#)), the MIT-licensed Ballycanew phosphorus tools, the CC BY records of the BNMA repository ([BNMA 2026](#)), the CC BY networks collected in bnRep ([Leonelli 2025](#)));
- fetch-only: metadata only. No-derivatives, non-commercial and unstated licences are never committed and never converted; fetch\_model downloads them into a scratch cache when a direct URL exists, and import\_model copies a file you downloaded by hand;
- builtin: built in Julia (the reference models of the SPEC);
- reconstruction: a data or code archive rather than a network. Nothing is committed and nothing is cached; load\_model raises ReconstructionOnlyError pointing at the record.

```
model_catalog(kind = :package)
```

ModelCatalog with 5 models

| name                            | category    | format  | licence                                                                                         |
|---------------------------------|-------------|---------|-------------------------------------------------------------------------------------------------|
| redistribution nodes arcs kind  |             |         |                                                                                                 |
| -----                           | -----       | -----   |                                                                                                 |
| -----                           | -----       | -----   |                                                                                                 |
| seagrass_dbn_package            | habitat     | package | Dryad record terms (consult the record; Dryad lists its content as reusable) reconstruction 0 0 |
| thailand_vertibrate_bbn_package | habitat     | package | figshare collection terms (consult the individual items) reconstruction 0 0                     |
| vansjo_seasonal_bn_package      | water       | package | unstated (repository states no licence; check the repository terms) reconstruction 0 0          |
| wild_pig_hsi_package            | biosecurity | package | CC0 1.0                                                                                         |
| bnrep_package                   | benchmark   | package | MIT                                                                                             |

```
try
  load_model("bnrep_package")
catch e
  println(sprint(showerror, e))
end
```

ReconstructionOnlyError: model "bnrep\_package" is a catalogue record for a data or code archive, not a network file, so there is nothing to load. The archive is at <https://github.com/manueleleonelli/bnRep>. Licence: MIT. Rebuild the network from the archive yourself, then register it as a model of its own with `import_model`.

```
model_catalog(redistributable = false)
```

ModelCatalog with 26 models

| name                                             | category   | format         | licence                       |
|--------------------------------------------------|------------|----------------|-------------------------------|
| redistribution nodes arcs kind                   |            |                |                               |
| -----                                            | -----      | -----          |                               |
| -----                                            | -----      | -----          |                               |
| beach_mice_bn                                    | habitat    | neta           | CC0 1.0 (ScienceBase record); |
| US Government work in the public domain          |            | fetch-only     | 0 0 BN                        |
| brown_bear_recreation                            | habitat    | dne            | CC-BY-NC-ND (version          |
| unstated)                                        |            |                | fetch-only 37 45              |
| BN                                               |            |                |                               |
| gulf_sturgeon_bn                                 | habitat    | neta           | CC0 1.0 (ScienceBase record); |
| US Government work in the public domain          |            | fetch-only     | 0 0 BN                        |
| plexus_teb_bat_site                              | habitat    | dne            | unstated                      |
| fetch-only 13 15 BN                              |            |                |                               |
| plexus_teb_bat_subwatershed                      | habitat    | dne            | unstated                      |
| fetch-only 7 6 BN                                |            |                |                               |
| seagrass_dbn_package                             | habitat    | package        | Dryad record terms (consult   |
| the record; Dryad lists its content as reusable) |            | reconstruction | 0 0                           |
| package                                          |            |                |                               |
| thailand_vertibrate_bbn_package                  | habitat    | package        | figshare collection terms     |
| (consult the individual items)                   |            | reconstruction | 0 0                           |
| package                                          |            |                |                               |
| bull_trout_food_web                              | population | dne            | CC-BY-NC-ND (version          |
| unstated)                                        |            |                | fetch-only 10 15              |
| BN                                               |            |                |                               |
| marten_age                                       | population | dne            | CC-BY (version unstated)      |
| fetch-only 7 6 BN                                |            |                |                               |
| marten_telomere                                  | population | dne            | CC-BY-NC-ND (version          |
| unstated)                                        |            |                | fetch-only 0 0                |
| BN                                               |            |                |                               |
| pacific_walrus                                   | population | dne            | CC-BY-NC-ND (version          |
| unstated)                                        |            |                | fetch-only 77 96              |
| BN                                               |            |                |                               |
| plexus_population_viability                      | population | dne            | unstated                      |
| fetch-only 11 13 ID                              |            |                |                               |
| plexus_teb_bat_basin                             | population | dne            | unstated                      |
| fetch-only 8 7 BN                                |            |                |                               |

|                                         |             |                |                             |    |    |
|-----------------------------------------|-------------|----------------|-----------------------------|----|----|
| polar_bear_stressor_i                   | population  | dne            | CC-BY-NC-ND (version        |    |    |
| unstated)                               |             |                | fetch-only                  | 38 | 44 |
| BN                                      |             |                |                             |    |    |
| polar_bear_stressor_ii                  | population  | dne            | CC-BY-NC-ND (version        |    |    |
| unstated)                               |             |                | fetch-only                  | 49 | 53 |
| BN                                      |             |                |                             |    |    |
| vansjo_seasonal_bn_package              | water       | package        | unstated (repository states |    |    |
| no licence; check the repository terms) |             | reconstruction | 0                           | 0  |    |
| package                                 |             |                |                             |    |    |
| shark_management_survey                 | risk        | dne            | CC-BY-NC (version unstated) |    |    |
| fetch-only 26 54 BN                     |             |                |                             |    |    |
| tidal_saline_wetlands                   | risk        | dne            | CC-BY-ND (version unstated) |    |    |
| fetch-only 18 16 BN                     |             |                |                             |    |    |
| arhopalus_flight_activity               | biosecurity | dne            | CC-BY-NC-ND (version        |    |    |
| unstated)                               |             |                | fetch-only                  | 7  | 11 |
| BN                                      |             |                |                             |    |    |
| fisram_freshwater_species               | biosecurity | dne            | CC-BY-ND (version unstated) |    |    |
| fetch-only 20 23 BN                     |             |                |                             |    |    |
| hylastes_flight_activity                | biosecurity | dne            | CC-BY-NC-ND (version        |    |    |
| unstated)                               |             |                | fetch-only                  | 7  | 11 |
| BN                                      |             |                |                             |    |    |
| hylurgus_flight_activity                | biosecurity | dne            | CC-BY-NC-ND (version        |    |    |
| unstated)                               |             |                | fetch-only                  | 10 | 17 |
| BN                                      |             |                |                             |    |    |
| microctonus_ng_risk                     | biosecurity | xdsl           | CC-BY-NC-ND (version        |    |    |
| unstated)                               |             |                | fetch-only                  | 45 | 51 |
| BN                                      |             |                |                             |    |    |
| wild_pig_hsi_package                    | biosecurity | package        | CC0 1.0                     |    |    |
| reconstruction 0 0 package              | benchmark   | dne            | CC-BY-NC-ND (version        |    |    |
| animals                                 |             |                | fetch-only                  | 7  | 6  |
| unstated)                               |             |                |                             |    |    |
| BN                                      |             |                |                             |    |    |
| bnrep_package                           | benchmark   | package        | MIT                         |    |    |
| reconstruction 0 0 package              |             |                |                             |    |    |

Every verbatim model carries its licence text:

```
print(licence_text("water"))
```

This network file is redistributed verbatim from the bnlearn Bayesian Network Repository (<https://www.bnlearn.com/bnrepository/>), maintained by Marco Scutari, under the Creative Commons Attribution-Share Alike 3.0 Unported licence (CC BY-SA 3.0, <https://creativecommons.org/licenses/by-sa/3.0/>).

Attribution: Marco Scutari, bnlearn Bayesian Network Repository.  
Scutari, M. (2010). Learning Bayesian Networks with the bnlearn R Package. Journal of Statistical Software, 35(3), 1-22. <https://doi.org/10.18637/jss.v035.i03>

The file is unmodified. Derivative works must be distributed under the same licence.

## 5 Loading a benchmark network

water (Jensen et al. 1989, from the bnlearn repository ([Scutari 2010](#))) is stored gzipped exactly as served; the loader decompresses it in memory, applies the manifest's parser options and hands the NetworkIR to BayesianNetworks.BayesModel.

```
water = load_model("water")
```

```
BayesModel(32 variables, 32 mechanisms, 32 kernels)
```

```
using BayesianNetworks: validate
validate(water; semantics = true)
```

model\_summary parses a model and counts what is in it without building a BayesModel, so it also works for influence diagrams:

```
model_summary("water")
```

```
ModelSummary water (WATER: waste-water treatment plant control)
  format / licence      net / CC BY-SA 3.0 (verbatim)
  nodes                 32 (32 chance, 0 decision, 0 utility)
  arcs                  66
  largest state space   4
  largest in-degree     5
  load_model            BayesModel
```

```
model_summary("waterhole_fence")
```

```
ModelSummary waterhole_fence (Waterhole Fence: fencing and plant survival decision
network)
  format / licence      dne / CC-BY (version unstated) (verbatim)
  nodes                 10 (6 chance, 1 decision, 3 utility)
  arcs                  12
  largest state space   3
  largest in-degree     3
  load_model            InfluenceDiagramModel
```

Decision models load as InfluenceDiagrams.InfluenceDiagramModels, with their chance kernels and tabular utilities bound and the decision nodes' parents as information sets:

```
koalas = load_model("koalas")
```

```
InfluenceDiagramModel(6 variables, 2 decisions, 4 utilities, 4 kernels, 4 bound)
```

```
using InfluenceDiagrams: optimize, policy_table
sol = optimize(koalas)
sol.expected_utility, policy_table(sol.strategy[:Cull]),
policy_table(sol.strategy[:Relocate])
```

```
(20.199999999999996, fill(:No), [:Yes, :Yes, :Yes])
```

The NetworkIR behind any model is one call away:

```
[(v.id, v.kind, length(v.parents)) for v in model_ir("koalas").variables]
```

```
10-element Vector{Tuple{Symbol, BayesianNetworkFormats.NodeKind, Int64}}:
 (:Cull, BayesianNetworkFormats.DecisionNode, 0)
 (:Relocate, BayesianNetworkFormats.DecisionNode, 1)
 (:KoalaPopulation, BayesianNetworkFormats.ChanceNode, 2)
 (:U, BayesianNetworkFormats.UtilityNode, 1)
 (:NveSocialImplications, BayesianNetworkFormats.ChanceNode, 2)
 (:V, BayesianNetworkFormats.UtilityNode, 1)
 (:TreeSpecies, BayesianNetworkFormats.ChanceNode, 1)
 (:Tourists, BayesianNetworkFormats.ChanceNode, 1)
 (:W, BayesianNetworkFormats.UtilityNode, 1)
 (:X, BayesianNetworkFormats.UtilityNode, 1)
```

## 6 A marginal on a small network

native\_fish\_v1 (Bayesian Intelligence, CC BY) links rainfall, drought, pesticide use and river flow to native fish abundance. With seven nodes the reference joint distribution is tiny, so BayesianNetworks.marginal answers exactly.

```
using BayesianNetworks: marginal, observe, probability
fish = load_model("native_fish_v1")
marginal(fish, :FishAbundance)
```

```
FiniteKernel{Float64}(I → FishAbundance{High,Medium,Low})
()
  High    0.2569
  Medium  0.2239
  Low     0.5193
```

Conditioning on a drought year:

```
marginal(observe(fish, :Drought ⇒ :Yes), :FishAbundance)
```

```
FiniteKernel{Float64}(I → FishAbundance{High,Medium,Low})
()
  High    0.1288
  Medium  0.1756
  Low     0.6957
```

```
[(s, round(probability(marginal(observe(fish, :Drought ⇒ :Yes), :FishAbundance),
s); digits = 3))
 for s in (:High, :Medium, :Low)]
```

```
3-element Vector{Tuple{Symbol, Float64}}:
 (:High, 0.129)
 (:Medium, 0.176)
 (:Low, 0.696)
```

## 7 The fetch-only workflow

Fetch-only models have a `download_url` (when the host serves the file directly) and `fetch_instructions` for the manual route. Nothing below touches the network: the calls that would be shown, not run.

```
spec = model_info("tidal_saline_wetlands")
(spec.licence, spec.redistribution, spec.download_url)

("CC-BY-ND (version unstated)", "fetch-only",
 "https://bnma.co/bnrepo/?slice=download&format=dne&bnId=209")

println(spec.fetch_instructions)
```

Open the BNMA export page (`source_url` with `body=convert`), choose 'Netica .dne format', press Save, and then `import_model(name, path)`; or call `fetch_model(name)`, which uses the same GET endpoint the Save button uses. The licence forbids redistribution of the file, so it is never committed.

Asking for the file before it is fetched raises a `ModelNotFetchedError` whose message repeats the instructions. Rendering this vignette must not disturb whatever you have already downloaded, so the demonstration runs inside `with_cache_dir`, which points `cache_dir()` at a temporary directory for the duration of the block. The error message names that directory by its absolute path, which differs from machine to machine, so the vignette prints it as `<temporary cache>`:

```
mktempdir() do tmp
  with_cache_dir(tmp) do
    try
      model_path("tidal_saline_wetlands")
    catch e
      println(replace(sprint(showerror, e), cache_dir() => "<temporary
        cache>"))
    end
  end
end
```

`ModelNotFetchedError`: the file of model "tidal\_saline\_wetlands" is not in the cache (<temporary cache>). Run `fetch_model("tidal_saline_wetlands")` to download it from <https://bnma.co/bnrepo/?slice=download&format=dne&bnId=209>, or obtain it by hand and call `import_model("tidal_saline_wetlands", path)`. Instructions: Open the BNMA export page (`source_url` with `body=convert`), choose 'Netica .dne format', press Save, and then `import_model(name, path)`; or call `fetch_model(name)`, which uses the same GET endpoint the Save button uses. The licence forbids redistribution of the file, so it is never committed.

`with_cache_dir` is the safe way to run anything that writes to or clears the cache (`fetch_model`, `import_model`, `clear_cache`); setting `ECOLOGICAL_BN_CACHE` does the same for a whole session. Nothing above deleted a file, and nothing below does either: the two ways to make the model available are shown, not run.

```
fetch_model("tidal_saline_wetlands")           # direct download, checksum verified
import_model("tidal_saline_wetlands", "Tidal Saline Wetlands.dne") # a file saved by
hand
clear_cache(name = "tidal_saline_wetlands")     # remove it again
```

After a fetch, `is_available("tidal_saline_wetlands")` is true and `model_ir` reads the file from the cache (`cache_dir()`). `verify_checksums(include_cache = true)` re-checks every committed and cached file against the manifests, and `verify_checksums(collect_mismatches = true)` reports every bad file rather than stopping at the first.

```
is_available("tidal_saline_wetlands"), length(fetchable_models())
```

```
(true, 20)
```

## 8 Reference models

Two models are built in Julia rather than read from files: the SPEC section 45 habitat network (`reference_habitat_bn`, structure only, and `reference_habitat_model`, with its conditional probability tables) and the section 46 grazing-management influence diagram (`reference_grazing_id`, an `InfluenceDiagramModel`). The following vignettes work through both.

```
habitat = load_model("reference_habitat_bn")
marginal(habitat, :Occupancy)
```

```
FiniteKernel{Float64}{I → Occupancy{absent,present}}
  ()
  absent  0.5238
  present 0.4762
```

```
reference_grazing_id()
```

```
InfluenceDiagramModel(10 variables, 1 decision, 2 utilities, 9 kernels, 2 bound)
```

## 9 Summary

The zoo is a registry of manifests: every model records its source, citation, DOI, licence, checksum and retrieval date, and the `redistribution` field decides whether the file is committed verbatim, fetched into a scratch cache on demand, built in Julia, or carried as a catalogue record only. That policy is what makes it safe to ship published networks alongside the code, and `licence_text` reproduces the terms next to every verbatim file. The next vignette, *The reference habitat network*, works one model end to end.

## 10 References

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